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Bai, Linxue; Pournik, Omid; Ding, Xuefei; Abbasi, Saadullah Farooq; Zhang, Le; Arvanitis, Theodoros

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Opening the Personal Gate between Technology and Health Care
M. Giacomini et al. (Eds.)

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Membership Inference or Data Split Bias? Identifying False Positives in Synthetic Medical Image Privacy Audits

Linxue BAI[a], Omid POURNIK[a], Xuefei DING[a], Saadullah Farooq ABBASI[a], Le ZHANG[a], and Theodoros ARVANITIS[a,1]
[a] *University of Birmingham, UK*
ORCiD ID: Linxue Bai https://orcid.org/0009-0008-0538-2307, Omid Pournik https://orcid.org/0000-0001-7938-0269,Xuefei Ding https://orcid.org/0009-0001-3073-9514, Saadullah Farooq Abbasi https://orcid.org/0000-0001-9814-3023, Le Zhang https://orcid.org/0000-0002-3848-0017, Theodoros Arvanitis https://orcid.org/0000-0001-5473-135X

Abstract. Membership Inference Attacks (MIAs) are the current standard for auditing the privacy of synthetic medical data. A high attack score is widely interpreted as a significant privacy leak, but this interpretation assumes the MIA is learning a true membership signal. We argue that these attacks may instead be learning experimental artefacts, specifically data split bias. To test this, we conduct experiments on synthetic brain MRI images from a StyleGAN3 model. We design a Primary MIA (member vs. non-member data) and a critical Control Experiment (disjoint subsets of member data alone). This member-only setup lacks any genuine membership distinction between classes, leaving only data split bias. Our results show the Control Experiment's AUCs match or exceed the Primary MIA's (+0.0205 on the 5k-pool Small corpus and +0.0113 on the 30k-pool Big corpus). This compelling evidence demonstrates that a high MIA score can be a false positive for a privacy breach, as the classifier is simply learning statistical artefacts. We conclude that MIA results are unreliable without rigorous control experiments to validate that they are detecting genuine privacy risks, not experimental bias.

Keywords. Membership Inference Attack (MIA), Synthetic Medical Images, Generative Models, Data Split Bias, Privacy Auditing, False Positives.

1. Introduction

Generative models, such as Generative Adversarial Networks [1] (GANs), have shown immense promise for creating realistic synthetic medical images, potentially revolutionising data sharing for research without exposing sensitive patient information [2, 3]. However, the "syntheticity" of this data is not an automatic guarantee of privacy [4]. A sophisticated generative model might inadvertently "memorise" and leak information about the specific individuals in its training set [5].

[1] Corresponding Author: Professor Theodoros N. Arvanitis, Department of Electronic, Electrical and Systems Engineering, School of Engineering, College of Engineering and Physical Sciences, University of Birmingham, Edgbaston, Birmingham, B15 2TT, United Kingdom. T.Arvanitis@bham.ac.uk

Despite the overall under-consideration of privacy risks compared to fidelity and utility, Membership Inference Attacks (MIAs) [6] have emerged as the de facto standard for privacy auditing where it is performed [7]. Established MIA methods against generative models include loss-based attacks using a model's output [8, 9], distance-based attacks measuring proximity [10, 11], and reconstruction-based attacks [11].

A high MIA accuracy is typically interpreted as a significant privacy breach. However, this interpretation has been challenged. Recent studies provide evidence that high MIA accuracy can be an artefact of the experimental setup, such as learning distributional differences between training and test data, rather than genuine privacy leakage [12]. Overestimating privacy risk can unnecessarily hinder safe data sharing under healthcare governance.

Furthermore, the standard validation paradigms for MIAs do not isolate this core issue. Typically, an attack's success is demonstrated on a relative basis: either by showing its accuracy is high on an unprotected model and low on a privacy-protected model (e.g., one trained with Differential Privacy) [13], or by showing a new attack method outperforms an old one [11]. Crucially, neither approach proves that the measured signal represents true privacy leakage rather than a statistical artefact.

This raises a critical question: Are we simply measuring the artefacts, not the privacy breach? In this paper, we investigate a specific artefact: data split bias. We hypothesise that powerful classifiers can distinguish between arbitrary data partitions even without a membership signal. We test this on a StyleGAN3 model by supplementing the standard MIA with a novel Control Experiment, where the attack model distinguishes between disjoint subsets of training data alone. Our findings show high attack accuracy even in this "member-only" setting, providing strong evidence that the model is learning the split bias. To the best of our knowledge, this is the first study to empirically isolate this confounding factor in medical image synthesis, demonstrating that current auditing tools can produce "false positive" privacy alerts. We conclude that rigorous control experiments are essential to validate genuine privacy risks.

2. Methodology

To investigate if MIAs learn data split bias, we designed a Primary MIA and a critical Control Experiment, both trained on high-similarity real-synth image pairs (Figure 1).

Our dataset consists of ~95.5k 2D MRI slices from the BraTS2023 Challenge I [14] (pre-processing applied). This data was partitioned into a training set, D_{source} (~91.2k slices from 1,190 patient cases, used to train a StyleGAN3 model [15]), and a held-out set, $D_{heldout}$ (~4.3k slices from 56 patient cases). We allocated a maximal portion to training to satisfy the data-hungry StyleGAN3 architecture; crucially, our Control Experiment (detailed later in this section) relies exclusively on internal splits of D_{source}, ensuring that data-split artefacts arise independently of the held-out set size.

Our Primary MIA tests if a classifier can distinguish members from non-members by focusing specifically on the most ambiguous cases. We focused on the T1-ce modality. For **real images**, the non-member pool was $D_{heldout}$ consisting of 4,334 images. The member pool D_{rep} was a comparable, representative subset of 4,615 images created by sampling 5% of the slices from each patient in D_{source}. This ensures the two real image pools are of balanced size, preventing a trivial bias in the nearest neighbour search.

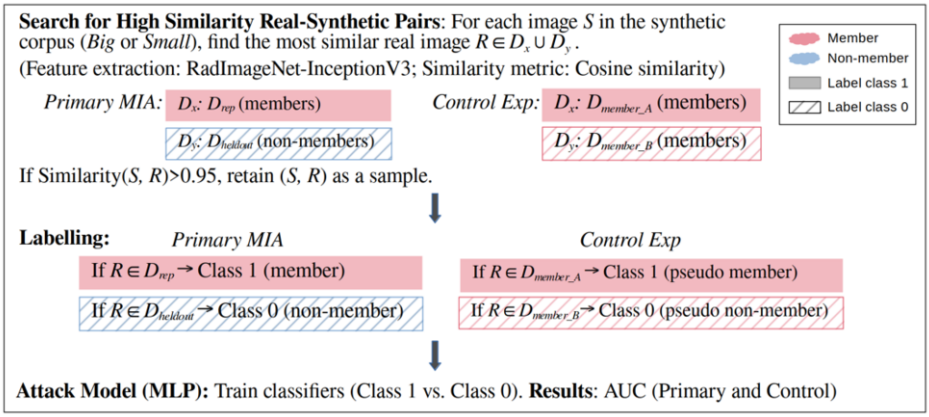


Figure 1. Experimental Design for the Primary Membership Inference Attack (MIA) and Control Experiment.

For **synthetic images**, we assessed robustness across two data scales: one using a search pool of 30,000 synthetic images and the other using 5,000. These resulting datasets are referred to as the Big Corpus and Small Corpus in this paper. All images were encoded using a pre-trained RadImageNet-InceptionV3 network [16]. Following the process in Figure 1, we filtered pairs based on the similarity and balanced the classes, yielding our final datasets of ~820 (Small) and ~5,060 (Big) pairs.

To isolate data split bias, we designed a Control Experiment acting as a statistical negative control. While standard MIAs distinguish members (training set) from non-members (held-out), this Control setup operates exclusively on members. Thus, any classification performance above random guessing (50%) reflects experimental artefacts rather than genuine privacy leakage. This establishes a baseline to isolate non-privacy-related factors essential for validating the primary MIA.

The Control Experiment used the same pipeline as the Primary MIA (Figure 1) but with different data sources. We randomly selected two small, disjoint groups of patients from D_{source} and used all their slices to create the member-only subsets, D_{member_A} and D_{member_B} (~4.5k images each). Pairs were deceptively labelled Class 0 ("pseudo non-members") or Class 1 ("pseudo members") based on their subset origin. To ensure a fair comparison, the final Control Experiment dataset was downsampled where needed to closely match the pair count of the Primary MIA dataset.

An identical multi-layer perceptron (MLP, 2 hidden layers) was used for all conditions. The MLP was trained on the concatenated feature vectors of the synthetic-real image pair to predict the label associated with the real image's origin. For each run, we used an 80/20 train/test split. We report average AUCs and ROC curves over 25 runs (5 distinct paired datasets × 5 model seeds; each paired dataset was generated from a new random sample of 5k or 30k synthetic images). We also applied a bootstrap analysis (N=10,000 resamples) for significance tests.

3. Results

Table 1 summarises the average AUCs for our Primary MIA and Control Experiment. The Primary MIA achieved a high AUC (0.8216 on the Small and 0.9655 on the Big corpus), which would typically suggest a privacy leak. However, the Control Experiment,

which operates in a member-only setting, performed better (0.8421 on the Small and 0.9768 on the Big corpus). The 95% CIs show that the confidence intervals for the difference (Control - Primary) are entirely above zero. This provides strong evidence that the Control Experiment reliably outperforms the Primary MIA.

Table 1. Average AUC (± standard deviation) for Primary MIA and Control Experiment (N=25 runs).

Metric	Small Corpus	Big Corpus
Primary MIA Avg. AUC (±Std)	$0.8216(\pm 0.0210)$	$0.9655(\pm 0.0056)$
Control Exp Avg. AUC (±Std)	$0.8421(\pm 0.0335)$	$0.9768(\pm 0.0062)$
Difference (Control−Primary)	0.0205	0.0114
95% CI of Difference	$[0.0052, 0.0359]$	$[0.0081, 0.0146]$

This key finding is visualised in Figure 2, which plots the averaged ROC curves and standard deviations for all four cases. The plot clearly shows two patterns: i) the solid lines (Big Corpus) are higher than the dashed lines (Small Corpus), as expected when using more data, and ii) in both cases, the red line (Control) is consistently above the blue line (Primary MIA).

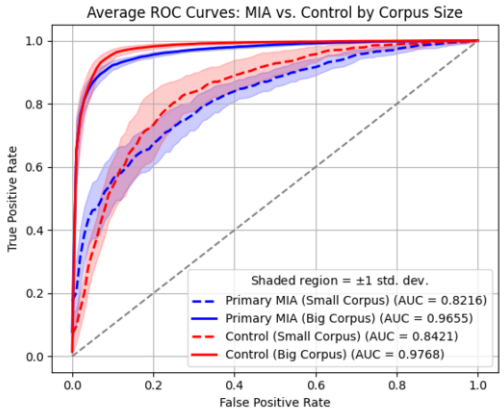


Figure 2. Average ROC curves (shaded standard deviation) for Primary MIA (blue) and Control (red) at both Small (dashed) and Big (solid) corpus scales.

4. Discussion and Conclusion

Our results show the Control Experiment's AUC consistently exceeded that of the Primary MIA at both small (0.8421 vs. 0.8216) and large (0.9768 vs. 0.9655) data scales. Achieving near-perfect accuracy in a member-only setting—where no genuine membership difference exists—demonstrates that data split bias is a dominant, learnable signal that overshadows the weaker, noisier "membership" signal. While prior studies suggest MIAs exploit distributional differences [12], our work provides the first direct empirical demonstration of this in medical image synthesis. Unlike comparisons between protected and unprotected models, our member-only control isolates data-split bias as a standalone confounder. The exceptionally high Control AUCs confirm these scores stem from structural artefacts rather than privacy loss.

These findings expose a vulnerability in current privacy assessment protocols. Without rigorous controls, MIAs risk systematically overestimating leakage, producing false positives that misclassify secure data as unsafe. Consequently, we recommend

incorporating negative control experiments into future regulatory guidelines to resolve the ambiguity between genuine risks and statistical quirks. Beyond privacy, these findings caution against "false confidence" in broader medical machine learning. Similar data-split artefacts may inflate performance in diagnostic models if patient-level correlations are preserved across splits. Rigorous negative controls are thus essential to prevent interpreting experimental artefacts as genuine model efficacy.

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